

1.	hacksmith@Notebook:~/Downloads\$	r2 -w	"libnative_functions.so"
	■	radare2 is WYSIWYF - what you see is what you fix	
2.	[0x000100b0]>	aaa	
3.	INFO: Analyze all flags starting with sym. and entry0 (aa)		
4.	INFO: Analyze imports (af@@@i)		
5.	INFO: Analyze all functions arguments/locals (afva@@@F)		
6.	INFO: Analyze function calls (aac)		
7.	INFO: Analyze len bytes of instructions for references (aar)		
8.	INFO: Finding and parsing C++ vtables (avrr)		
9.	INFO: Finding function preludes (aap)		
10.	INFO: Finding xrefs in noncode section (e anal.in=io.maps.x)		
11.	INFO: Analyze value pointers (aav)		
12.	INFO: aav: 0x00000000-0x00034f3c in 0x0-0x34f3c		
13.	INFO: Emulate functions to find computed references (aaef)		
14.	INFO: Type matching analysis for all functions (aافت)		
15.	INFO: Propagate noreturn information (aanr)		
16.	INFO: Use -AA or aaaa to perform additional experimental analysis		
17.	[0x000100b0]>	afl	
18.	0x0000fbe0	1	16 sym.imp.realloc
19.	0x0000fc00	1	16 sym.imp.pthread_key_create
20.	0x0000fc10	1	16 sym.imp.__vsprintf_chk
21.	0x0000fc20	1	16 sym.imp.pthread_once
22.	0x0000fc30	1	16 sym.imp.__cxa_finalize
23.	0x0000fc50	1	16 sym.imp.calloc
24.	0x0000fc60	1	16 sym.imp.syslog
25.	0x0000fc80	1	16 sym.imp.posix_memalign
26.	0x0000fc90	1	16 sym.imp.read
27.	0x0000fca0	1	16 sym.imp.__strlen_chk
28.	0x0000fcc0	1	16 sym.imp.abort
29.	0x0000fce0	1	16 sym.imp.__errno
30.	0x0000fcf0	1	16 sym.imp.dl_iterate_phdr
31.	0x0000fd00	1	16 sym.imp.android_set_abort_message
32.	0x0000fd10	1	16 sym.imp.islower
33.	0x0000fd30	1	16 sym.imp.__memmove_chk

34.	0x0000fd40	1	16 sym.imp.__read_chk
35.	0x0000fd80	1	16 sym.imp.fputc
36.	0x0000fd90	1	16 sym.imp.__stack_chk_fail
37.	0x0000fda0	1	16 sym.imp.dlclose
38.	0x0000fdb0	1	16 sym.imp.pthread_key_delete
39.	0x0000fdc0	1	16 sym.imp.pthread_cond_wait
40.	0x0000fdd0	1	16 sym.imp.memchr
41.	0x0000fde0	1	16 sym.imp.pthread_setspecific
42.	0x0000fdf0	1	16 sym.imp.dlsym
43.	0x0000fe00	1	16 sym.imp.openlog
44.	0x0000fe20	1	16 sym.imp.dlerror
45.	0x0000fe60	1	16 sym.imp.strcmp
46.	0x0000fe70	1	16 sym.imp.__open_2
47.	0x0000fe90	1	16 sym.imp.syscall
48.	0x0000fed0	1	16 sym.imp.pthread_mutex_lock
49.	0x0000fee0	1	16 sym.imp.closelog
50.	0x0000fef0	1	16 sym.imp.pthread_getspecific
51.	0x0000ff10	1	16 sym.imp.memmove
52.	0x0000ff50	1	16 sym.imp.isxdigit
53.	0x0000ff90	1	16 sym.imp.strlen
54.	0x0000ffa0	1	16 sym.imp.exit
55.	0x0000ffb0	1	16 sym.imp.close
56.	0x0000ffc0	1	16 sym.imp.vasprintf
57.	0x0000ffd0	1	16 sym.imp.dlopen
58.	0x00010000	1	16 sym.imp.malloc
59.	0x00010010	1	16 sym.imp.memcpy
60.	0x00010020	1	16 sym.imp.memset
61.	0x00010040	1	16 sym.imp.pthread_cond_broadcast
62.	0x00010060	1	16 sym.imp.free
63.	0x00010070	1	16 sym.imp.vfprintf
64.	0x00010080	1	16 sym.imp.pthread_mutex_unlock
65.	0x00010090	1	16 sym.imp.__cxa_atexit
66.	0x000100b0	1	12 entry0
67.	0x00015a48	6	120 sym.__cxa_call_unexpected
68.	0x000146c4	1	24 sym.__cxa_free_exception
69.	0x000160d4	16	288 fcn.000160d4

409.	0x0002867c	8	196	method._anonymous_namespace_::itanium_demangle::ElaboratedTypeSpefType.virtual_32
410.	0x00028768	22	460	method._anonymous_namespace_::itanium_demangle::PointerType.virtual_32
411.	0x00028934	17	300	method._anonymous_namespace_::itanium_demangle::PointerType.virtual_40
412.	0x00028c34	13	256	method._anonymous_namespace_::itanium_demangle::ReferenceType.virtual_40
413.	0x00028d48	1	56	method._anonymous_namespace_::itanium_demangle::PostfixQualifiedType.virtual_32
414.	0x0002924c	1	40	fcn.0002924c
415.	0x0002b100	1	144	fcn.0002b100
416.	0x0002b3fc	57	1052	fcn.0002b3fc
417.	0x0002b818	1	156	fcn.0002b818
418.	0x0002c0b8	7	156	fcn.0002c0b8
419.	0x0002c160	4	76	fcn.0002c160
420.	0x0002c1a0	5	148	fcn.0002c1a0
421.	0x0002c240	2	56	fcn.0002c240
422.	0x0002c370	3	36	fcn.0002c370
423.	0x0002c558	3	64	fcn.0002c558
424.	0x0002c598	8	128	fcn.0002c598
425.	0x000003b4	1	4	fcn.000003b4
426.	0x00013c88	1	16	method.j51_13u_.virtual_0
427.	0x00013c98	1	8	method.j51_13u_.virtual_8
428.	0x00000400	1	4	fcn.00000400
429.	[0x000100b0]> axt @0x0000ffa0			
430.	entry.init0	0x13594	[CALL:--x] bl sym.imp.exit	
431.	[0x000100b0]> 0x13594			
432.	[0x00013594]> pdr			
433.	;-- Java_io_ymusic_nativelib_impl_NativeUtil_init:			
434.	; NULL XREF from aav.0x00001f0c @ +0x102c(r)			

435.	/ 164: entry.init0 (int64_t arg1, int64_t arg2, int64_t arg_30h);		
436.	; arg int64_t arg1 @ x0		
437.	; arg int64_t arg2 @ x1		
438.	; arg int64_t arg_30h @ sp+0x60		
439.	; var int64_t var_8h @ sp+0x8		
440.	; var int64_t var_10h @ sp+0x10		
441.	; var int64_t var_10h_2 @ sp+0x18		
442.	; var int64_t var_20h @ sp+0x20		
443.	; var int64_t var_20h_2 @ sp+0x28		
444.	0x000134f4	ffc300d1	sub sp, sp, 0x30
445.	0x000134f8	f44f01a9	stp x20, x19, [var_10h]
446.	0x000134fc	fd7b02a9	stp x29, x30, [var_20h]
447.	0x00013500	fd830091	add x29, var_20h
448.	0x00013504	480100b0	adrp x8, section..data ; 0x3c000
449.	0x00013508	08c10b91	add x8, x8, 0x2f0
450.	0x0001350c	08fddf08	ldarb w8, [x8]
451.	0x00013510	f30300aa	mov x19, x0 ; arg1
452.	0x00013514	540100b0	adrp x20, section..data ; 0x3c000
453.	0x00013518	28020036	tbz w8, 0, 0x1355c
454.	// true: 0x0001355c false: 0x0001351c		
455.	; CODE XREFS from entry.init0 @ 0x13570(x), 0x1358c(x)		
456.	0x0001351c	817641f9	ldr x1, [x20, 0x2e8] ; pstate ; int64_t arg2
457.	0x00013520	e00313aa	mov x0, x19 ; int64_t arg1
458.	0x00013524	71faff97	bl fcn.00011ee8
459.	0x00013528	817641f9	ldr x1, [x20, 0x2e8] ; pstate ; int64_t arg2
460.	0x0001352c	f403002a	mov w20, w0
461.	0x00013530	e00313aa	mov x0, x19 ; int64_t arg1
462.	0x00013534	1df8ff97	bl fcn.000115a8
463.	0x00013538	a8939c52	mov w8, 0xe49d
464.	0x0001353c	2857a872	movk w8, 0x42b9, lsl 16

465.		0x00013540	1f00086b	cmp w0, w8	
466.		0x00013544	61020054	b.ne 0x13590	
467.		// true: 0x00013590 false: 0x00013548			
468.		0x00013548	54020037	tbnz w20, 0, 0x13590	
469.		// true: 0x00013590 false: 0x0001354c			
470.		0x0001354c	fd7b42a9	ldp x29, x30, [var_20h]	
471.		0x00013550	f44f41a9	ldp x20, x19, [var_10h]	
472.		0x00013554	ffc30091	add sp, arg_30h	
473.		0x00013558	c0035fd6	ret	
474.		; CODE XREF from entry.init0 @ 0x13518(x)			
475.		0x0001355c	400100b0	adrp x0, section..data	
			; 0x3c000		
476.		0x00013560	00c00b91	add x0, x0, 0x2f0	
477.		0x00013564	e10700f9	str x1, [var_8h]	;
		arg2			
478.		0x00013568	cef2ff97	bl fcn.000100a0	
479.		0x0001356c	e10740f9	ldr x1, [var_8h]	;
		pstate			
480.		0x00013570	60fdff34	cbz w0, 0x1351c	
481.		// true: 0x0001351c false: 0x00013574			
482.		0x00013574	e00313aa	mov x0, x19	;
		int64_t arg1			
483.		0x00013578	16f4ff97	bl fcn.000105d0	
484.		0x0001357c	807601f9	str x0, [x20, 0x2e8]	
485.		0x00013580	400100b0	adrp x0, section..data	
		; 0x3c000			
486.		0x00013584	00c00b91	add x0, x0, 0x2f0	
487.		0x00013588	46f2ff97	bl fcn.0000fea0	
488.		0x0001358c	e4ffff17	b 0x1351c	
489.		// true: 0x0001351c			
490.		; CODE XREFS from entry.init0 @ 0x13544(x), 0x13548(x)			
491.		0x00013590	e0031f2a	mov w0, wzr	
492.	\	0x00013594	83f2ff97	bl sym.imp.exit	;
		void exit(int status)			
493.		[0x00013594]> 0x00013518			

494.	[0x00013518]> wa b 0x0001351c
495.	INFO: Written 4 byte(s) (b 0x0001351c) = wx 01000014 @ 0x00013518
496.	[0x00013518]> 0x00013544
497.	[0x00013544]> wa b 0x00013548
498.	INFO: Written 4 byte(s) (b 0x00013548) = wx 01000014 @ 0x00013544
499.	[0x00013544]> 0x00013548
500.	[0x00013548]> wa b 0x0001354c
501.	INFO: Written 4 byte(s) (b 0x0001354c) = wx 01000014 @ 0x00013548
502.	[0x00013548]>